

Yuanchen (Cathy) Li

(415)509-9350

yuanchen@uchicago.edu

<https://cathyliyuanchen.github.io/>

Research Interests

Machine Learning, Artificial Intelligence, Security and Privacy, Cryptography

Education

University of Chicago

2023.9 - Present

Ph.D. in Computer Science.

Awards: *Neubauer Fellowship*.

University of California, Berkeley

2017.8 - 2020.12

B.A. in Computer Science, Statistics, Applied Mathematics, Cognitive Science, Linguistics, and Data Science.

Awards: *Magna cum laude (GPA: 3.88/4.00). Upsilon Pi Epsilon. Phi Beta Kappa.*

Publications

1. “Understanding Implosion in Text-to-Image Generative Models”
Wenxin Ding, Cathy Y. Li, Shawn Shan, Ben Y. Zhao, Haitao Zheng
Proceedings of the ACM Conference on Computer and Communications Security (CCS), Oct 2024.
 2. “Inception Attacks: Immersive Hijacking in Virtual Reality”
Zhuolin Yang, Cathy Y. Li, Arman Bhalla, Ben Y. Zhao, Haitao Zheng
In submission.
 3. “SALSA Fresca: Angular Embeddings and Pre-Training for ML Attacks on LWE”
Samuel Stevens, Emily Wenger, Cathy Y. Li, Eshika Saxena, François Charton, Kristin Lauter
Transactions on Machine Learning Research (TMLR), 2025.
 4. “The Cool and the Cruel: Separating Hard Parts of LWE Secrets”
Niklas Nolte*, Mohamed Malhou*, Emily Wenger*, Samuel Stevens, Cathy Y. Li, François Charton, Kristin Lauter
AFRICACRYPT, July 2024.
 5. “An efficient algorithm for integer lattice reduction”
François Charton, Kristin Lauter, Cathy Y. Li, Mark Tygert
SIAM Journal on Matrix Analysis and Applications (SIMAX), 45 (1): 353-367, 2024.
 6. “SALSA Verde: a machine learning attack on Learning With Errors with sparse small secrets”
Cathy Y. Li, Emily Wenger, Zeyuan Allen-Zhu, François Charton, Kristin Lauter
Proceedings of the 37th Conference on Neural Information Processing Systems (NeurIPS), Nov 2023.
 7. “SALSA Picante: a machine learning attack on LWE with binary secrets”
Cathy Y. Li, Jana Sotáková, Emily Wenger, Mohamed Malhou, Evrard Garcelon, François Charton, Kristin Lauter
Proceedings of the ACM Conference on Computer and Communications Security (CCS), Nov 2023.
-

Employment

Ph.D. Student, University of Chicago

2023.9 - Present

Advisor: *Prof. Grant Ho*

- Leading a research project on adversarial attacks on large language models for cybersecurity.
- Proposed a theoretical framework to understand poison attacks on generative models, with empirical verifications on latent diffusion models.
- Proposed a multifaceted defense pipeline on an immersive hijacking attack on virtual reality systems.

Ph.D. Intern, Nvidia

2025.6 - 2025.9

- Worked on XR robot teleoperation, adding support to the robot simulation framework (Isaac Sim and Isaac Lab).
- Enabled hand tracking with Manus gloves and Vive trackers, and lower body control with foot pedals on Isaac Lab.

AI Resident, Meta

2022.7 - 2023.7

Mentors: Prof. Kristin Lauter and Dr. François Charton

- Main contributor of the cryptanalysis with ML project. Proposed and implemented innovative ideas to improve data preprocessing, model training, and secret recovery algorithms.
- Scaled up the attack to solve LWE problems of larger dimensions (from 128 to 512), smaller modulus, and broader secret type. Improved the attack time to outperform state-of-the-art.
- Invited to speak at JMM 2025. The work has inspired subsequent studies in AI4Crypt.

Software Engineer, Productiv

2021.6 - 2022.7

- Built a workflow engine to standardize workflows. Wrote backend APIs and shipped features in the product that allows customers to automatically off-board employees, and customize license management configurations according to app engagement data. Presented a demo in the company all-hands.
- Provided timely technical solutions to customer inquiries as the backend on-call of the engineering team.

Research Assistant, Real-time Intelligent Secure Explainable Systems Lab, UC Berkeley

2019.5 - 2021.5

Mentor: Prof. Joseph Gonzalez

- Worked with PhD students on Computer Vision and Machine Learning projects, and presented in lab seminars.
- Observed spatial bias in data and sparsity of feature maps and demonstrated a tradeoff between restriction for a convolution and accuracy on a segmentation model. Proposed leveraging spatial biases to save computation.
- Designed a CNN to predict diagnosis from electrocardiograms (EKG) signals; implemented the neural network from scratch and trained on a large scale EKG dataset from UCSF, visualizing the attention.

Teaching**Teaching Assistant, University of Chicago**

2025.9 - 2026.3

*CMSC 25300: Mathematical Foundations of Machine Learning**CMSC 25400: Machine Learning***Teaching Assistant, University of Chicago**

2024.1 - 2024.3

*CMSC 23400: Mobile Computing***(Head) Undergraduate Student Instructor, UC Berkeley**

2019.8 - 2020.12

CS 188: Artificial Intelligence, an upper division course with 800 enrolled students

- 4 Semesters as (Head) TA. Led staff of TAs and readers; assisted in course logistics and administrative responsibilities; organized staff meetings and mentored new TAs; worked on content (notes and assignments) creation.
- Organized exam creation/proctoring/grading; created exam questions; made and revised exam drafts.
- Held discussions and office hours, and answered student questions on the online Q&A platform.

Skills

Proficient in Python (Pytorch, NumPy, Pandas, Scikit-learn, Matplotlib), Node.js, R (ggplot2), SQL, Unix, Git, Latex. Spoken languages: Mandarin (native), English (working proficiency), Swedish (elementary).